

Yuanfan Chen

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EDUCATION

Cornell Tech, Cornell University
Master of Engineering in Computer Science

New York, NY
Aug. 2025 – May. 2026

University of Toronto
Honours Bachelor of Science, Computer Science Specialist (GPA: 3.87/4.0)

Toronto, Canada
Sep. 2020 – May. 2024

EXPERIENCE

Machine Learning System Researcher
Cornell University

Sep 2025 – Present
New York, NY

- Quantified that reasoning workloads cause **50% output-length misprediction** in LLM request schedulers; demonstrated via simulation that this misrouting degrades **P99 TTFT by up to 8×**, revealing a fundamental gap in production serving systems.
- Developed **PTA** (Prefill→Think→Answer) three-stage disaggregated serving; identified that each phase impacts UX differently, enabling **differential SLO constraints**; independently configured **DVFS frequency**, **KV-Cache allocation**, and **parallelism** per stage — preserving user-perceivable latency (**TTFAT**) via Answer-phase frequency protection under power caps while improving overall throughput through stage-level resource tuning.
- Built a quantitative **GPU frequency–performance model** by systematically profiling **DVFS** frequency vs. power, throughput, and latency curves on A100/H100; designed an **online optimization framework** that jointly schedules **LC/Flex/BE** request classes, maximizing throughput and QoS under **instantaneous power caps** and **cumulative energy budget** constraints.
- Applied **γ -Boost Queuing Theory** to design a prediction-free, tail-aware scheduler with **KV-cache-aware preemption**; reduced **tail TTFT/TTLT by 37–60%** while improving throughput by **1.01–1.12×** on Azure production traces.

Machine Learning Engineer
Tencent: Distributed Training and Inference

May 2025 – Jul 2025
Beijing, CN

- Designed **request-level admission control** with **batch-size stepwise profiling** for latency estimation and **SLA-aware priority scheduling**; maintained **100% goodput** under **2–10×** **QPS bursts** by enforcing per-class latency budgets across mixed-priority traffic.
- Identified **high prefix redundancy** in production traffic and adopted **SGLang with RadixAttention** for prefix caching; implemented **prefill-first chunking** over decode-first, reducing **P99 TTFT by ~40%** without sacrificing decode throughput.
- Optimized CUDA attention kernels for **GQA** (A100) and **DeepSeek FlashMLA** sparse attention (H200); profiled with **Nsight Compute** to localize memory-hierarchy bottlenecks and redesigned thread-block tiling, shared-memory layout, and warp scheduling; achieved **~20% throughput gain** on GQA and **~15%** on FlashMLA over baselines.

Kernel Development Intern
Tencent: TCHouse-X Distributed Database

Jul 2025 – Aug 2025
Beijing, CN

- Designed and implemented an **admission control mechanism** to regulate request concurrency, ensuring system stability under bursty loads (which later inspired my Multi-SLO research).
- Contributed to the open-source **ClickHouse-X** project; established rigorous **CI/CD pipelines** and code review standards to build robust, maintainable, and reproducible distributed systems.
- Optimized node recovery via fault-tolerant **KV delta-apply**, reducing latency from **10s to 3s** and ensuring data consistency in high-churn clusters.

Systems Researcher
Far Data Lab

Jan 2024 – Feb 2025
Toronto, CA

- Co-designed **dpBento**, a modular benchmark framework that standardizes performance evaluation across heterogeneous DPUs (**BlueField-2/3**, **OCTEON TX2**), from micro-level primitives to end-to-end cloud database offloading.
- Profiled **RocksDB** compaction and decompression paths end-to-end; offloaded compute-heavy stages to DPU, reducing **host CPU utilization by ~40%** and achieving **~3×** **decompression throughput** via hardware acceleration.

TECHNICAL SKILLS

Programming: Python, C++, CUDA, Java **AI Infra:** vLLM, SGLang, KV-Cache, PD separation
Distributed: Raft, ClickHouse, HDFS **Databases:** RocksDB, MySQL, Redis
Networking: epoll, multiplexing, async I/O, KCP **Systems:** Linux internals, CPU/GPU arch, Nsys/Nsight

PUBLICATIONS

- Making Sense of DPU Performance for Cloud Data Processing: [Experiment, Analysis & Benchmark]**. Jiasheng Hu, Chihan Cui, Yuanfan Chen, Philip A. Bernstein, Jialin Li, Qizhen Zhang. arXiv:2504.05536 (2025).

ACADEMIC SERVICE

- Invited Reviewer:** IEEE Journal on Selected Areas in Information Theory (JSAIT) – Special Issue on Energy and Data Efficiency in AI (2026). Reviewed submissions focusing on cutting-edge LLM inference systems.